

## RF Exposure Evaluation Report

**Product** : Seeed Studio XIAO ESP32S3,  
Seeed Studio XIAO ESP32S3 Sense

**Trade mark** : Seeed Studio

**Model/Type reference** : XIAO-ESP32-S3, XIAO-ESP32-S3-Sense

**Serial Number** : N/A

**Report Number** : EED32P80440603

**FCC ID** : Z4T-XIAOESP32S3

**Date of Issue** : Apr. 20, 2023

**Test Standards** : 47 CFR Part 1.1307  
47 CFR Part 1.1310  
47 CFR Part 2.1091  
47 CFR Part 2.1093  
447498 D04 Interim General RF Exposure  
Guidance v01

**Test result** : PASS

Prepared for:

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Prepared by:

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Version

| Version No. | Date          | Description |
|-------------|---------------|-------------|
| 00          | Apr. 20, 2023 | Original    |
|             |               |             |
|             |               |             |

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### 3 General Information

#### 3.1 Client Information

|                          |                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------|
| Applicant:               | Seeed Technology Co., Ltd                                                                                   |
| Address of Applicant:    | 9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.                    |
| Manufacturer:            | Seeed Technology Co., Ltd                                                                                   |
| Address of Manufacturer: | 9F, Building G3, TCL International E city, Zhongshanyuan Road, Nanshan, Shenzhen, China.                    |
| Factory:                 | Shenzhen Xinxian Technology Co., Limited                                                                    |
| Address of Factory:      | F5, Building B17, Hengfeng Industrial City, No. 739 Zhoushi Rd, Baoan District, Shenzhen, Guangdong, P.R.C. |

#### 3.2 General Description of EUT

|                 |                                                            |
|-----------------|------------------------------------------------------------|
| Product Name:   | Seeed Studio XIAO ESP32S3, Seeed Studio XIAO ESP32S3 Sense |
| Model No.(EUT): | XIAO-ESP32-S3, XIAO-ESP32-S3-Sense                         |
| Trade Mark:     | Seeed Studio                                               |

#### 3.3 Product Specification subjective to this standard

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Frequency Range:      | BLE: 2402MHz~2480MHz<br>2.4G WIFI: IEEE 802.11b/g/n(HT20): 2412MHz to 2462MHz<br>IEEE 802.11n(HT40): 2422MHz to 2452MHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| Modulation Type:      | BLE: GFSK<br>2.4G WIFI: IEEE for 802.11b: DSSS(CCK,DQPSK,DBPSK)<br>IEEE for 802.11g :OFDM(64QAM, 16QAM, QPSK, BPSK)<br>IEEE for 802.11n(HT20 and HT40) : OFDM (64QAM, 16QAM,QPSK,BPSK)                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| Test Power Grade:     | Default                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| Test Software of EUT: | ESP32S3.exe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| Antenna Type:         | FPC antenna                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| Antenna Gain:         | 2.90dBi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| Power Supply:         | USB port:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | DC 5.0V |
| Sample Received Date: | Mar. 31, 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
| Sample tested Date:   | Mar. 31, 2023 to Apr. 11, 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |
| Remark:               | <p>Company Name and Address shown on Report, the sample(s) and sample Information was/ were provided by the applicant who should be responsible for the authenticity which CTI hasn't verified.</p> <p>Model No.: XIAO-ESP32-S3, XIAO-ESP32-S3-Sense</p> <p>XIAO-ESP32-S3-Sense is composed of XIAO-ESP32-S3 main board and expansion board. XIAO-ESP32-S3 is a single main board without an expansion board, and compared with the above-mentioned XIAO-ESP32-S3 Sense main board, there is no connector on the board. Their electrical circuit design, layout, components used and internal wiring are identical.</p> |         |

### 3.4 Test Location

All tests were performed at:

Centre Testing International Group Co., Ltd

Building C, Hongwei Industrial Park Block 70, Bao'an District, Shenzhen, China

Telephone: +86 (0) 755 33683668 Fax: +86 (0) 755 33683385

No tests were sub-contracted.

FCC Designation No.: CN1164

### 3.5 Deviation from Standards

None.

### 3.6 Abnormalities from Standard Conditions

None.

### 3.7 Other Information Requested by the Customer

None.



## 4 SAR Evaluation

### 4.1 RF Exposure Compliance Requirement

#### 4.1.1 Limits

The SAR-based exemption formula of § 1.1307(b)(3)(i)(B), repeated here as Formula (B.2), applies for single fixed, mobile, and portable RF sources with available maximum time-averaged power or effective radiated power (ERP), whichever is greater, of less than or equal to the threshold  $P_{th}$  (mW).

This method shall only be used at separation distances from 0.5 cm to 40 cm and at frequencies from 0.3 GHz to 6 GHz (inclusive).  $P_{th}$  is given by Formula

$$P_{th} \text{ (mW)} = \begin{cases} ERP_{20 \text{ cm}} (d/20 \text{ cm})^x & d \leq 20 \text{ cm} \\ ERP_{20 \text{ cm}} & 20 \text{ cm} < d \leq 40 \text{ cm} \end{cases}$$

where

$$x = -\log_{10} \left( \frac{60}{ERP_{20 \text{ cm}} \sqrt{f}} \right)$$

and  $f$  is in GHz,  $d$  is the separation distance (cm), and  $ERP_{20 \text{ cm}}$  is per Formula (B.1).

$$P_{th} \text{ (mW)} = ERP_{20 \text{ cm}} \text{ (mW)} = \begin{cases} 2040f & 0.3 \text{ GHz} \leq f < 1.5 \text{ GHz} \\ 3060 & 1.5 \text{ GHz} \leq f \leq 6 \text{ GHz} \end{cases} \quad (\text{B.1})$$

The 1 mW Blanket Exemption of § 1.1307(b)(3)(i)(A) applies for single fixed, mobile, and portable RF sources with available maximum time-averaged power of no more than 1 mW, regardless of separation distance.

#### 4.1.2 Test Procedure

Software provided by client enabled the EUT to transmit and receive data at lowest, middle and highest channel individually.

**4.1.3 EUT RF Exposure Evaluation****For Stand alone:****For BLE**

| Frequency (MHz) | Max. Conducted Output power (dBm) | Antenna Gain (dBi) | EIRP (dBm) | ERP (dBm) | ERP (mW) | Limit (mW) | Result |
|-----------------|-----------------------------------|--------------------|------------|-----------|----------|------------|--------|
| 2480            | 4.82                              | 2.90               | 7.72       | 5.57      | 3.606    | ≤3060      | PASS   |

**For 2.4G Wi-Fi**

| Frequency (MHz) | Max. Conducted Output power (dBm) | Antenna Gain (dBi) | EIRP (dBm) | ERP (dBm) | ERP (mW) | Limit (mW) | Result |
|-----------------|-----------------------------------|--------------------|------------|-----------|----------|------------|--------|
| 2462            | 18.05                             | 2.90               | 20.95      | 18.80     | 75.858   | ≤3060      | PASS   |

**Note:**

- ① EIRP=conducted power+antenna gain;
- ② ERP=EIRP-2.15
- ③ Only the worst case data was recorded in the report.

The test report is effective only with both signature and specialized stamp, The result(s) shown in this report refer only to the sample(s) tested. Without written approval of CTI, this report can't be reproduced except in full.

**\*\*\* End of Report \*\*\***